

Jayden Shin

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Software Engineering

- **Relevant Courses:** Programming Principles, Linear Algebra, Calculus 1, Introduction to Methods of Software Engineering

Oakridge Secondary School

London, ON

Ontario Secondary School Diploma

- 5s on AP Calculus BC, Physics C: Mechanics, Statistics, Macroeconomics, and Computer Science: A; **1580 SAT**

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Dart, Python, C++, Rust

Frameworks & Tools: Flutter, React, Node.js, Angular, NumPy, PyTorch

EXPERIENCE

Paradym Private Capital | London, ON

Apr. 2026 - Present

Head of Technology Operations

- Operated end-to-end technology operations for a private capital firm managing **\$70K+ in assets**; engineered algorithmic and web-based systems that contributed to **20%+ annualized returns** in collaboration with an Ivey AEO and Queens Commerce executive team

Adopt Our Community | London, ON

Apr. 2025 - Jun. 2026

Volunteer IT Director

- Built internal web platforms and automated data-routing systems across **10+ chapters**; coordinated with **19+ external partners** to streamline logistics and drive **\$10K+ in fundraising outcomes**

United Nations | New York City, NY

Mar. 2025 - Apr. 2025

Youth Representative

- Selected as one of **~100 global youth representatives** to the United Nations alongside PhD candidates and government ministers; delivered a live, televised policy address to **1,000+ attendees**, including Council President Bob Rae

Free Your Mind Initiative | London, ON

Mar. 2025 - Nov. 2025

Web Developer

- Designed and delivered a responsive CMS web platform within a 5-member team to modernize a non-profit's digital operations; assumed site-administration ownership post-launch, maintaining **continuous uptime and content accuracy**

RookBook Sports | London, ON

Jun. 2024 - Oct. 2024

Software Development Intern

- Built and maintained the production website for a Southeast Asian football management agency as a software development intern; mentored by senior engineers and developed hands-on proficiency in **Flutter-based full-stack development** across front-end and back-end systems

AWARDS

Best Start-Up and Prototype

Shad Canada

- 1st place in the Shad Design Competition (month-long start-up pitch judged by professionals) for a modular trailer skirt of HDPE and biomass foam to cut emissions and fuel costs

Semifinalist

Wix Creators of Tomorrow Challenge

- Top 22 of thousands for a website built for the STEM Thrivers non-profit (funded by the City of Vancouver); advanced to the May 2025 finals for a **\$9K USD** prize pool

2 Honor Roll, 7 Distinction, 5 School Champion

Waterloo Math Contests

- Honor roll (top 3%) on Galois and Hypatia; school champion and distinction (top 25%) on the Cayley, CIMC, and Fryer contests

PROJECTS

Laplace | Pokémon Showdown Battle AI — Python

[Demo](#) | [GitHub](#)

- Reached a peak **2137 Elo** — ranked **#447 in the world** on the competitive ladder, top 1% among hundreds of thousands of human players; drew **200K+ views** across social media

- Drives each turn with **determinized Monte Carlo Tree Search** to reason under hidden information and simultaneous moves: samples plausible opponent teams, simulates **~250K playouts / 120 ms** on a multithreaded Rust engine, and plays a mixed, unexploitable strategy
- Deduces concealed opponent items and stats from in-game tells (move patterns, turn order), and trained a **PyTorch value network** (368 features, self-play labels, ~70% accuracy) to resolve near-ties
- Validated every change through a loss-mining → root-cause → **A/B self-play harness (Wilson confidence intervals)**, benchmarked against the strongest open-source bot

Last Tour | *Girls' Last Tour New Tab Extension* — JavaScript

[Page](#) | [GitHub](#)

- Shipped a **Manifest V3** Chrome extension (200 impressions) that overrides the new-tab page with a procedural pixel-art city rendered to a low-res offscreen canvas and nearest-neighbour upscaled, with parallax layers and runtime snowfall
- Synthesized all ambience through the **WebAudio API** (filtered-noise wind, oscillator engine idle) and added a chrome.storage journal and topSites integration — shipping zero copyrighted assets

Progression Difficulty Scaler | *Minecraft Fabric Mod* — Java

[Page](#) | [GitHub](#)

- Built a server-side **Fabric** mod (**400 downloads**, MC 1.21.1) that hooks entity-load events to rescale each mob's max-health and attack-damage attributes on spawn against a server-wide progression average
- Derives a hidden **0-100 progression score** per player from gear, dimensions, advancements, and log-scaled diamond/netherite mining, mapped through a normalized logistic S-curve so difficulty ramps mid-to-late game and eases at the top